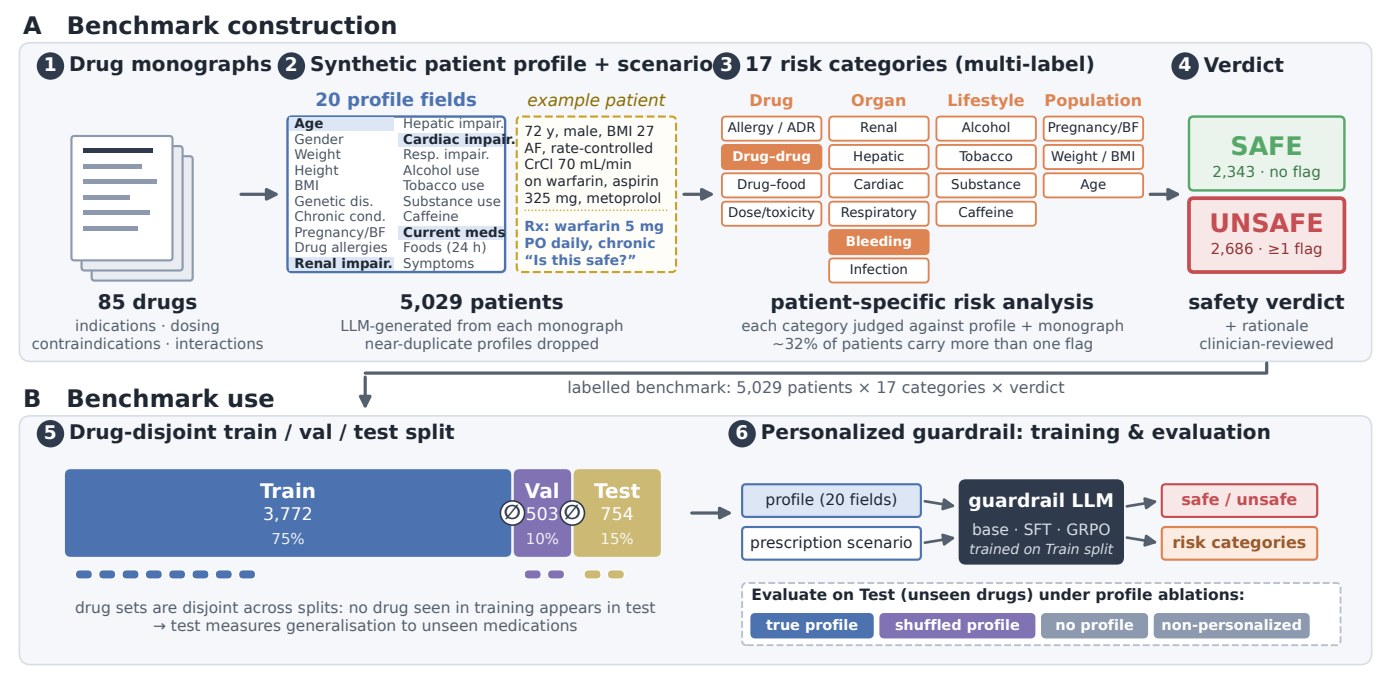
A Benchmark construction
1 Drug monographs
2 Synthetic patient profile + scenario
3 17 risk categories (multi-label)
4 Verdict
20 profile fields
example patient
Age
Gender
Weight
Height
BMI
Genetic dis.
Chronic cond.
Pregnancy/BF
Drug allergies
Renal impair.
Hepatic impair.
Cardiac impair.
Resp. impair.
Alcohol use
Tobacco use
Substance use
Caffeine
Current meds
Foods (24 h)
Symptoms
72 y, male, BMI 27
AF, rate-controlled
CrCl 70 mL/min
on warfarin, aspirin
325 mg, metoprolol
Rx: warfarin 5 mg
PO daily, chronic
"Is this safe?"
Drug
Allergy / ADR
Drug-drug
Drug–food
Dose/toxicity
Organ
Renal
Hepatic
Cardiac
Respiratory
Bleeding
Infection
Lifestyle
Alcohol
Tobacco
Substance
Caffeine
Population
Pregnancy/BF
Weight / BMI
Age
SAFE
2,343 · no flag
UNSAFE
2,686 · ≥1 flag
85 drugs
indications · dosing
contraindications · interactions
5,029 patients
LLM-generated from each monograph
near-duplicate profiles dropped
patient-specific risk analysis
each category judged against profile + monograph
~32% of patients carry more than one flag
safety verdict
+ rationale
clinician-reviewed
labelled benchmark: 5,029 patients × 17 categories × verdict
B Benchmark use
5 Drug-disjoint train / val / test split
Train
3,772
75%
Val
503
10%
Test
754
15%
drug sets are disjoint across splits: no drug seen in training appears in test
→ test measures generalisation to unseen medications
6 Personalized guardrail: training & evaluation
profile (20 fields)
prescription scenario
guardrail LLM
base · SFT · GRPO
trained on Train split
safe / unsafe
risk categories
Evaluate on Test (unseen drugs) under profile ablations:
true profile
shuffled profile
no profile
non-personalized